



PROJECT NAME: SCIENTIFIC UNIT CONVERTER

SUBMITTED BY:

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1ST SEMESTER

BTECH CSE (AI/ML)

SECTION: B

IBM

GUIDENCE/SUBMITTED TO:

MISS.NAINA DEVI

IBM

SESSION:2025-26

AGENDA

-  To design a menu-driven unit conversion program using C
- To perform temperature conversions
-  To convert distance units
-  To convert mass units
-  To convert time units
-  To demonstrate the use of switch– case and nested logic

OBJECTIVE

1.  To understand the structure of a C program
2.  To implement **switch– case statements** effectively
3.  To take user input and process it correctly
4. To apply logical and mathematical operations
5.  To provide accurate and fast unit conversions
6.  To enhance problem-solving and coding skills



INTRODUCTION TO PROJECT

Scientific Unit Converter Using c program

-  **Overview**
The Scientific Unit Converter is a console-based C application designed to convert different scientific units in an easy and efficient manner.
-  **Purpose**
This project helps users convert units related to **Temperature, Distance, Mass, and Time** using simple menu options. It also helps students understand practical implementation of switch– case statements in C programming.
-  **Working Principle**
The program first asks the user to select a category, then a conversion type, and finally the input value. Based on the selection, the program performs the required calculation and displays the result.
-  **Application**
This type of converter is useful in education, laboratories, engineering calculations, and daily-life unit conversions.



TOOLS & TECHNOLOGIES USED



Visual Studio Code (VS Code)

Used for writing and editing the C source code.



GCC Compiler

Compiles and executes the C program.



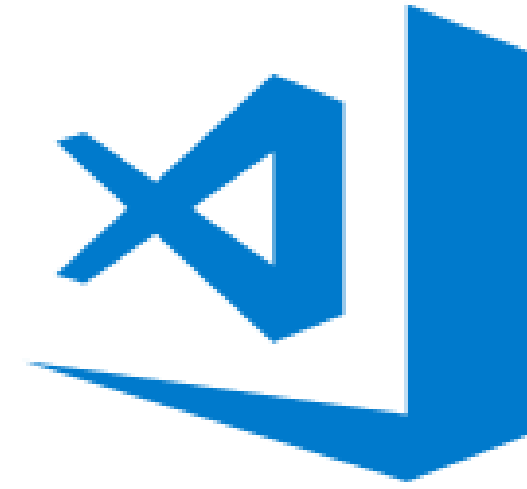
C Programming Language

Used to implement logic, conditions, and calculations.



Windows Operating System

Provides the execution environment.



Visual Studio Code



Windows Operating System

PROJECT WORKING

- 1   **Category Selection**
User selects a category like Temperature, Distance, Mass, or Time.
- 2   **Conversion Choice**
User chooses a specific conversion type from the menu.
- 3   **Input Value**
User enters the value to be converted.
- 4   **Logic Execution**
The program uses switch– case statements to perform calculations.
- 5   **Result Display**
Converted value is displayed with accuracy.

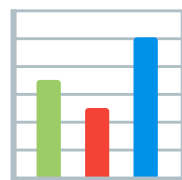
?? SCREENSHOT

```
new.c
C: > Users > Ujjwal Goswami > OneDrive > Desktop > c > c project > new.c

3  int main() {
4      int category, choice;
5      double input, output;
6
7      printf("--- Scientific Unit Converter ---\n");
8      printf("Select Category: \n1. Temperature\n2. Distance\n3. Mass\n4. Time\n");
9      printf("Enter choice (1-4): ");
10     scanf("%d", &category);
11
12     switch (category) {
13     case 1:
14         printf("\n1. Celsius to Fahrenheit\n2. Fahrenheit to Celsius\n3. Celsius to Kelvin\n");
15         printf("Enter choice: ");
16         scanf("%d", &choice);
17         printf("Enter value: ");
18         scanf("%lf", &input);
19
20         switch (choice) {
21             case 1: output = (input * 9/5) + 32; break;
22             case 2: output = (input - 32) * 5/9; break;
23             case 3: output = input + 273.15; break;
24             default: printf("Invalid Sub-choice"); return 1;
25         }
26         printf("Result: %.2lf\n", output);
27         break;
28
29     case 2:
30         printf("\n1. Meters to Kilometers\n2. Miles to Kilometers\n3. Feet to Meters\n");
31         printf("Enter choice: ");
32         scanf("%d", &choice);
33         printf("Enter value: ");
34         scanf("%lf", &input);
35
36         switch (choice) {
37             case 1: output = input / 1000; break;
38             case 2: output = input * 1.60934; break;
39             case 3: output = input * 0.3048; break;
40             default: printf("Invalid Sub-choice"); return 1;
41         }
42         printf("Result: %.2lf\n", output);
43     }
```

```
new.c
C: > Users > Ujjwal Goswami > OneDrive > Desktop > c > c project > new.c

3  int main() {
12     switch (category) {
29         case 2:
42             printf("Result: %.2lf\n", output);
43             break;
44
45         case 3:
46             printf("\n1. Kilograms to Pounds\n2. Grams to Kilograms\n");
47             printf("Enter choice: ");
48             scanf("%d", &choice);
49             printf("Enter value: ");
50             scanf("%lf", &input);
51
52             switch (choice) {
53                 case 1: output = input * 2.20462; break;
54                 case 2: output = input / 1000; break;
55                 default: printf("Invalid Sub-choice"); return 1;
56             }
57             printf("Result: %.2lf\n", output);
58             break;
59
60         case 4:
61             printf("\n1. Hours to Minutes\n2. Minutes to Seconds\n3. Days to Hours\n");
62             printf("Enter choice: ");
63             scanf("%d", &choice);
64             printf("Enter value: ");
65             scanf("%lf", &input);
66
67             switch (choice) {
68                 case 1: output = input * 60; break;
69                 case 2: output = input * 60; break;
70                 case 3: output = input * 24; break;
71                 default: printf("Invalid Sub-choice"); return 1;
72             }
73             printf("Result: %.2lf\n", output);
74             break;
75
76         default:
77             printf("Invalid Category Selected.\n");
78
79
80     return 0;
}
```



Output

[PROBLEMS](#)[OUTPUT](#)[DEBUG CONSOLE](#)[TERMINAL](#)[PORTS](#)[Code](#) + - [] [X] ... | [] [X]

```
PS C:\Users\Ujjwal Goswami> cd "c:\Users\Ujjwal Goswami\OneDrive\Desktop\c\c project\" ; if ($?) { gcc new.c -o new } ; if ($?) { .\new }
```

```
--- Scientific Unit Converter ---
```

```
Select Category:
```

1. Temperature
2. Distance
3. Mass
4. Time

```
Enter choice (1-4): 4
```

1. Hours to Minutes
2. Minutes to Seconds
3. Days to Hours

```
Enter choice: 1
```

```
Enter value: 2
```

```
Result: 120.00
```

```
PS C:\Users\Ujjwal Goswami\OneDrive\Desktop\c\c project> cd "c:\Users\Ujjwal Goswami\OneDrive\Desktop\c\c project\" ; if ($?) { gcc new.c -o new } ; if ($?) { .\new }
```

```
--- Scientific Unit Converter ---
```

```
Select Category:
```

1. Temperature
2. Distance
3. Mass
4. Time

```
Enter choice (1-4): 1
```

1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Celsius to Kelvin

```
Enter choice: 2
```

```
Enter value: 3
```

```
Result: -16.11
```

```
PS C:\Users\Ujjwal Goswami\OneDrive\Desktop\c\c project> 
```

✓ Advantage of Scientific Unit Converter

- 1  Easy to Use 
Simple menu-driven interface.
- 2  Fast Conversion 
Instant results without manual calculation.
- 3  Accurate Results 
Reduces human calculation errors.
- 4  Educational 
Helps beginners learn C programming concepts.
- 5  Reusable 
Can be modified to add more conversions.



Future Scope of Scientific Unit Converter

- 1  Add More Units 
Area, Volume, Speed, Energy, etc.
- 2  GUI-Based Application 
Convert it into a graphical application.
- 3  File Handling 
Store conversion history in files.
- 4  Web or Mobile Version 
Develop as a web or mobile app.
- 5  Advanced Error Handling 
Improve validation for invalid inputs.

Thank

You